Fermi motion in untagged J/ψ photoproduction on the deuteron

Bremsstrahlung photon beam from the 5 cm target, spectator model of the deuteron, and the yield ratio to a free neutron at rest

V. Kubarovsky
Thomas Jefferson National Accelerator Facility

generator: `exclurad_py`, branch `jpsi-fermi`, module `generator/photo.py`

26 September 2026

Abstract

A 10.6 GeV electron beam radiates in the 5 cm liquid-deuterium cell of CLAS12 and the bremsstrahlung photons produce J/ψ mesons on the neutrons of the deuteron, $\gamma n \to J/\psi\, n$. We describe the photon spectrum used, including the correct behaviour at the endpoint $E_\gamma \to E_0$ where the complete-screening formula fails, the spectator model of the deuteron with the AV18 momentum distribution, and the Monte Carlo that combines them with the J/ψ cross section of the `roberts2026` parametrisation. With 20 million events per sample the ratio of the J/ψ yield per neutron on the deuteron to the yield on a free neutron at rest is measured to 0.2 % per 100 MeV bin of photon energy. Above $E_\gamma = 8.6$ GeV the ratio stays between 0.9889 and 1.0173, with a mean of 0.9944 ± 0.0003: Fermi motion changes the yield per neutron by less than 1 %. Two effects of comparable size and opposite sign make up this number: the binding of the struck neutron lowers the available γn energy and reduces the yield by about 4 %, and the Møller flux factor of the moving neutron, correlated with the steeply rising cross section, raises it by about the same amount. Fermi motion is therefore negligible for the yield above 8.6 GeV, but the shape of the ratio near the free threshold at 8.20 GeV and the yield below it, 2.55 % of the total, are entirely due to it.

Contents

1 Purpose and set-up **2**

2 The photon beam: bremsstrahlung in the target **2**
2.1 Thin-target spectrum . . . 2
2.2 The endpoint . . . 3

3 The deuteron: spectator model **4**
3.1 The event weight and the flux factor . . . 5

4 The elementary cross section **5**

5 The generator and its validation **6**
5.1 Algorithm . . . 6
5.2 Validation . . . 7

6 Results **7**
6.1 Yield versus photon energy and the ratio deuteron / free neutron . . . 7
6.2 The yield below the free threshold . . . 8
6.3 Spectator momenta and the off-shell mass . . . 9
6.4 The t distributions . . . 9

7 Discussion and limits **10**

A Cards, commands and files **10**

1 Purpose and set-up

The question asked was simple: how much does the Fermi motion of the neutron in the deuteron change the J/ψ photoproduction yield in the CLAS12 configuration, where the photon beam is the untagged bremsstrahlung of the 10.6 GeV electron beam in the target itself? The scattered electron is not observed; it stays in the beam line. The observable is the $J/\psi \to e^+e^-$ yield as a function of the photon energy reconstructed from the final state, and its t distribution.

The generator therefore has three ingredients, described in the next three sections:

1. the photon spectrum dN_γ/dE_γ per incident electron, from thin-target bremsstrahlung in the cell, with the endpoint treated correctly (Section 2);
2. the deuteron, in the plane-wave spectator model with the AV18 momentum distribution (Section 3);
3. the elementary cross section $\sigma(\gamma N \to J/\psi N)$ and its t dependence (Section 4).

Section 5 gives the algorithm and its validation, Section 6 the results of the high-statistics samples, and Section 7 the interpretation and the limits of the model. All samples were produced with `exclurad_py` on the `jpsi-fermi` branch; the cards and commands are in Appendix A.

Three samples are compared throughout, all with the same photon spectrum and the same elementary cross section:

sample	target	events
deuteron	neutron of the deuteron, AV18 Fermi motion, spectator proton written out	20000000
neutron	free neutron at rest	20000000
proton	free proton at rest (LH_2 radiator thickness), for cross-checks	20000000

The *deuteron/neutron* ratio, yield per neutron, isolates the effect of Fermi motion and binding; the photon flux and the cross section cancel in it bin by bin.

2 The photon beam: bremsstrahlung in the target

2.1 Thin-target spectrum

The beam electron of energy $E_0 = 10.6\,\text{GeV}$ traverses the liquid target and radiates. The 5 cm cell is $L/X_0 = 5/769.1 = 6.5 \times 10^{-3}$ radiation lengths of LD_2 (5.6×10^{-3} for LH_2, $X_0 = 890.4\,\text{cm}$), so the thin-target formula applies: the photon spectrum is linear in the thickness and the energy loss of the electron before it radiates is neglected (it is at most 1.7 MeV over the full cell, see Section 2.2). We use the complete-screening form of Tsai,

$$\frac{dN_\gamma}{dE_\gamma} = \frac{X}{X_0}\,\frac{1}{E_\gamma}\left[\tfrac{4}{3}\,(1-y)+y^2\right] S(E_\gamma), \qquad y = \frac{E_\gamma}{E_0}, \tag{1}$$

where X is the thickness of material the electron has crossed before the production vertex and $S(E_\gamma)$ is the endpoint factor of Section 2.2, equal to one except in the last $\sim$300 MeV. The definition of X_0 absorbs the atomic screening logarithms and the contribution of the atomic electron, which for hydrogen and deuterium is as large as that of the nucleus.

The photon flux grows linearly along the cell, so the production vertices are distributed with a density proportional to the upstream thickness, $dP/dz \propto z - z_{\rm in}$; the generator draws $z = z_{\rm in} + L\sqrt{u}$ with u uniform. Integrated over the cell, the yield per incident electron and per target nucleon is that of the average thickness, $X = L/2$, and this is the factor used in the event weight,

$$\frac{X}{X_0} = \frac{L}{2X_0} = 3.25 \times 10^{-3}\ (\mathrm{LD_2}), \qquad 2.81 \times 10^{-3}\ (\mathrm{LH_2}). \tag{2}$$

With these numbers the LD_2 cell delivers 1.439×10^{-3} photons per electron between 6.5 and 10.6 GeV, and 0.768×10^{-3} above the free-neutron threshold of 8.20 GeV (Figure 1, left).

2.2 The endpoint

Formula (1) without S is finite at $E_\gamma = E_0$: the bracket tends to one and the spectrum to $1/E_0$. This is an artefact of the complete-screening approximation, which assumes that the minimum momentum transfer to the atom,

$$q_{\rm min} = \frac{m_e^2\, E_\gamma}{2E\, E'}, \qquad E = E_0,\ E' = E_0 - E_\gamma, \tag{3}$$

is small compared with the inverse screening radius $1/a_s$ of the atomic electron cloud. As $E' \to 0$ the momentum transfer grows without bound, screening switches off, and the cross section is governed by the unscreened logarithm $\ln(2EE'/m_eE_\gamma)$, which falls through zero when E' becomes of the order of m_e. **The spectrum therefore does go to zero at the endpoint**, over the last fraction of an MeV, after bending down over the last few tens of MeV.

We use the analytic intermediate-screening spectrum of Schiff (Koch and Motz formula 3BS(b)), which interpolates between complete and no screening with the momentum transfer of each photon energy. With $r = E'/E$ and, in units of m_e,

$$b = \frac{2EE'}{a_s\, E_\gamma}, \qquad \frac{1}{M(0)} = \Big(\frac{E_\gamma}{2EE'}\Big)^2 + \frac{1}{a_s^2}, \tag{4}$$

Schiff's spectrum is

$$\begin{aligned}\frac{d\sigma}{dE_\gamma} \propto \frac{1}{E_\gamma}\Big\{&\Big[1 + r^2 - \tfrac{2}{3}r\Big]\Big[\ln M(0) + 1 - \tfrac{2}{b}\arctan b\Big] \\ &+ r\Big[\tfrac{2}{b^2}\ln(1+b^2) + \tfrac{4(2-b^2)}{3b^3}\arctan b - \tfrac{8}{3b^2} + \tfrac{2}{9}\Big]\Big\}.\end{aligned} \tag{5}$$

Its complete-screening limit $b \to \infty$ is $\frac{1}{E_\gamma}\{[1+r^2-\frac{2}{3}r](2\ln a_s+1)+\frac{2}{9}r\}$, and $1+r^2-\frac{2}{3}r = \frac{4}{3}(1-y)+y^2$ is exactly the Tsai bracket. The endpoint factor is the ratio

$$S(E_\gamma) = \frac{\text{Eq. (5)}}{\text{its complete-screening limit}}, \tag{6}$$

cut at zero where Schiff's bracket turns negative, and it multiplies the Tsai shape in Eq. (1). Applying the ratio rather than Eq. (5) itself keeps the normalisation of X_0, which contains the electron contribution and the Coulomb-free $Z = 1$ logarithms, and changes only the shape where the physics changes.

The screening radius is the one parameter. Schiff's Thomas–Fermi value $a_s = 111\, Z^{-1/3}$ (in Compton wavelengths) describes many-electron atoms; hydrogen and deuterium have a single $1s$ electron and the physical screening length is the Bohr radius, $a_0 = \lambda_C/\alpha = 137$ Compton wavelengths. We use 137; Figure 1 (right) shows that the difference from 111 is confined to the last few MeV. The resulting factor is

$E_0 - E_\gamma$ [MeV]	300	100	30	10	3	1
$S(E_\gamma)$	0.968	0.910	0.769	0.582	0.362	0.160

and $S = 0$ for $E_0 - E_\gamma < 0.42\,\mathrm{MeV}$. The endpoint correction removes 1.24 % of the photons between 6.5 and 10.6 GeV. Its effect on the J/ψ yield is proportionally larger than on the photon count, because the cross section is largest at the highest photon energies, but it is still a sub-percent effect on the total; what it does change is the last 100 MeV of the E_γ spectrum, which is what a measurement near the endpoint sees.

Three effects that we do not include act on the same scale. The electron loses up to 1.7 MeV by ionisation before it radiates ($dE/dx \simeq 0.35\,\mathrm{MeV/cm}$ in LD_2), the CEBAF beam has an energy spread of order 10^{-4}, i.e. 1 MeV, and the bremsstrahlung on the atomic electron has its own endpoint $1.5\,m_e$ below E_0. Each smears or shifts the tip by about 1 MeV; none matters for the J/ψ yield. In the last 5 MeV the formula is also outside its domain, $E' \gg m_e$; that region holds less than 10^{-4} of the photons above 6.5 GeV.

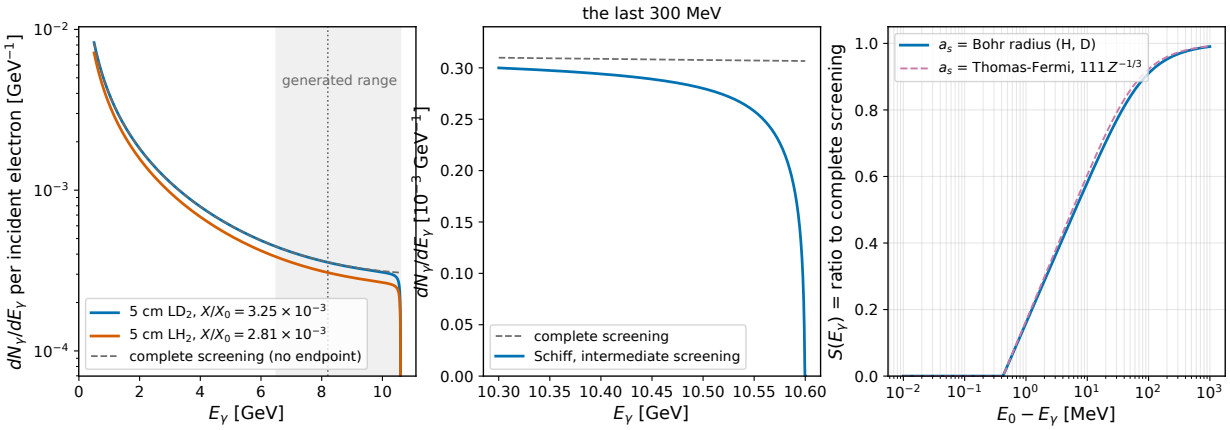


Figure 1: The photon beam. Left: photons per incident electron and per GeV from the 5 cm cell (average upstream thickness $L/2$), for LD_2 and LH_2; the dashed curve is the complete-screening formula without the endpoint factor; the shaded band is the generated range 6.5–10.6 GeV and the dotted line the free-neutron threshold. Middle: the last 300 MeV, where the Schiff intermediate screening spectrum bends down and reaches zero at the endpoint while the complete-screening formula stays finite. Right: the endpoint factor $S(E_\gamma)$ against the distance to the endpoint, for the Bohr radius used here and for Schiff's Thomas–Fermi radius.

3 The deuteron: spectator model

The deuteron is at rest in the laboratory. In the plane-wave spectator picture the photon interacts with one nucleon while the other is a spectator that leaves with the momentum it had in the nucleus. The spectator is on its mass shell, the struck nucleon carries the rest of the deuteron four-momentum and is off shell:

$$p_s = (E_s, \vec{p_s}), \quad E_s = \sqrt{M_p^2 + p_s^2}; \qquad p_N = (M_D - E_s, -\vec{p_s}), \qquad m^{*2} = p_N^2 < M_n^2. \tag{7}$$

The spectator momentum is drawn from the AV18 deuteron momentum distribution $n(p_s)$ (S and D waves, normalised to one, tabulated up to $p_s = 0.70\,\mathrm{GeV}/c$), isotropic in the laboratory. The photon–neutron invariant mass is built from the actual four-vectors,

$$s = (k + p_N)^2 = m^{*2} + 2E_\gamma\,(E_N - p_{N,z}), \tag{8}$$

so a neutron running toward the photon ($p_{N,z} < 0$) raises s and one running away lowers it. The free-neutron threshold $E_\gamma^{\mathrm{th}} = [(M_n + M_{J/\psi})^2 - M_n^2]/2M_n = 8.20\,\mathrm{GeV}$ is no longer a threshold: the reaction $\gamma d \to J/\psi\, n\, p$ is open from $[(M_{J/\psi} + M_n + M_p)^2 - M_D^2]/2M_D = 5.66\,\mathrm{GeV}$, and photons between 5.66 and 8.20 GeV produce J/ψ on the high-momentum tail of $n(p_s)$. The generated range starts at 6.5 GeV; below it the yield is negligible (Section 6.2).

Two conventions of the earlier electroproduction study are kept. The cross section is evaluated at the free-nucleon mass, as a function of the invariants s and t of the event, and m^* enters only where the event is built, i.e. in the kinematics of the final state. And a draw with $m^* < 0.10\,\mathrm{GeV}$ is rejected: there the struck nucleon becomes light-like and the plane-wave picture has no meaning. The floor was chosen for this study from the table of allowed configurations (Table 1): at 7.0 GeV, where only 0.9 % of the AV18 configurations can produce a J/ψ at all, the floor at 0.30 GeV used for π^0 would remove 3 % of them, the floor at 0.10 GeV removes 0.1 %.

Table 1: Kinematics of the struck neutron in the deuteron for a photon along $+z$ (AV18, 200 000 configurations, unweighted by the cross section): fraction of configurations above the J/ψ threshold, their mean spectator momentum, the fraction with $p_s > 300\,\mathrm{MeV}/c$, and the fraction of the allowed configurations removed by a floor on m^*.

E_γ [GeV]	allowed	$\langle p_s \rangle$ [MeV/c]	$p_s > 300$	floor 0.30	0.20	0.10
7.0	0.88 %	373	60.8 %	2.9 %	0.7 %	0.11 %
7.5	4.44 %	224	18.3 %	0.7 %	0.2 %	0.04 %
8.0	24.6 %	117	4.4 %	0.17 %	0.06 %	0.01 %
8.2	44.3 %	91	2.6 %	0.10 %	0.03 %	0.01 %
10.0	97.3 %	86	1.9 %	0.06 %	0.02 %	0.00 %

3.1 The event weight and the flux factor

Per incident electron and per target neutron, the number of J/ψ events is

$$dN = \frac{dN_\gamma}{dE_\gamma}\,dE_\gamma \;\times\; n(p_s)\,d^3p_s \;\times\; \frac{k\cdot p_N}{E_\gamma\,E_N}\,\sigma_{\gamma n}(s)\,, \qquad \frac{k\cdot p_N}{E_\gamma\,E_N} = 1-\beta_{N,z}, \tag{9}$$

where the last factor is the Møller flux factor of the moving target: the rate of a massless projectile on a target with velocity $\vec{\beta}_N$ is σ times the relative velocity $1-\beta_{N,z}$, not σ alone. For a nucleon at rest it is one, and averaged over the isotropic $n(p_s)$ it is also one. It is nevertheless not a detail here. A neutron running toward the photon has both a larger flux factor and a larger s, hence a larger cross section; the two are positively correlated and the average of the product exceeds the product of the averages. Because $\sigma_{\gamma n}(s)$ rises steeply above threshold (Section 4), this correlation is worth about +4 % on the deuteron yield at 10 GeV and more near threshold. It was absent from the first version of the generator, and its absence produced a spurious 4 % deficit of the deuteron yield; the samples in this report include it, and Section 6.1 shows the ratio with and without it. The same factor is missing in the electroproduction branch of the generator (`generator/fermi.py` working in the struck-nucleon rest frame), where its numerical effect is much smaller because the π^0 and η cross sections vary slowly with W; it should be added there for consistency.

4 The elementary cross section

The hadronic side is the electroproduction model `jpsi.roberts2026` at $Q^2 = 0$, relabelled for a neutron target as `jpsin.roberts2026` with $\sigma_n = \sigma_p$. Nothing new is assumed on this side: in the Hand convention the transverse cross section at $Q^2 = 0$ *is* the real-photon cross section, so the generator uses the model's $\sigma_T(W, Q^2{=}0, t)$ as $d\sigma/dt$. The parametrisation is

$$\sigma_{\mathrm{tot}}(W) = A\,(W - W_{\mathrm{th}})^{\alpha}\,e^{-B\,(W-W_{\mathrm{th}})}, \qquad \frac{d\sigma}{dt} = \sigma_{\mathrm{tot}}(W)\,\frac{b\,e^{b\,t'}}{1-e^{-b\,|t'|_{\max}}}, \quad t' = t - t_{\min}, \tag{10}$$

with $A = 1.018\,\mathrm{nb}$, $\alpha = 0.707$, $B = -0.928\,\mathrm{GeV}^{-1}$, $b = 1.67\,\mathrm{GeV}^{-2}$ and $W_{\mathrm{th}} = M_N + M_{J/\psi} = 4.036\,\mathrm{GeV}$. Between the free threshold and 10.6 GeV, $W - W_{\mathrm{th}}$ runs from 0 to 0.52 GeV and σ_{tot} from 0 to 1.05 nb (Figure 2); the logarithmic derivative $d\ln\sigma/dW = \alpha/(W-W_{\mathrm{th}}) - B$ is $2.7\,\mathrm{GeV}^{-1}$

at 10 GeV and diverges at threshold, which is what makes the deuteron yield sensitive to shifts of s of a few $10^{-2}\,\mathrm{GeV}^2$. The t range generated is $-6 \le t \le -0.001\,\mathrm{GeV}^2$, cut to the physical range at each W; $|t|_{\min}$ is 1.8, 0.62 and 0.36 GeV^2 at 8.5, 9.5 and 10.5 GeV.

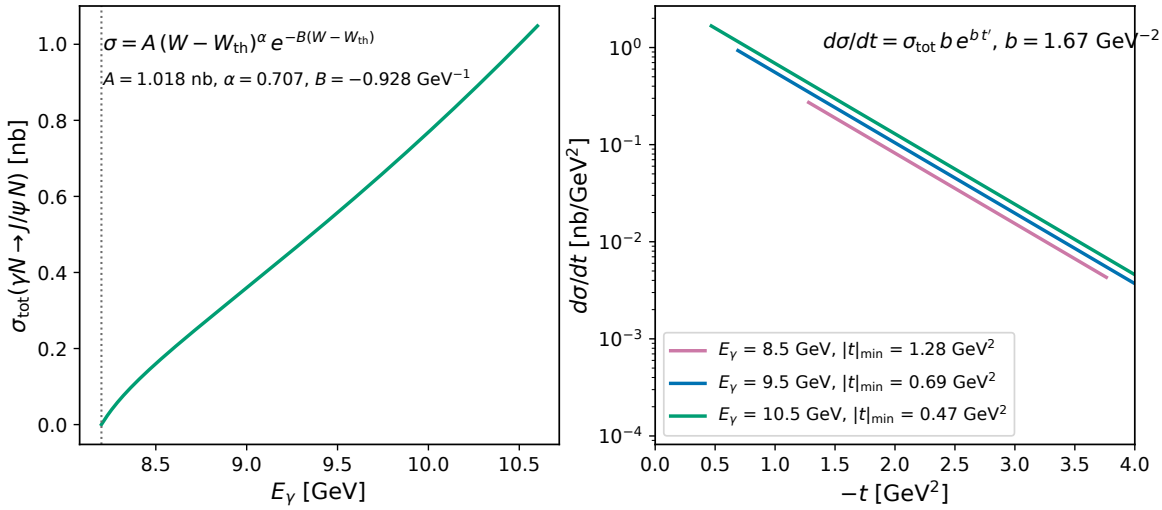


Figure 2: The elementary cross section used. Left: $\sigma_{\mathrm{tot}}(\gamma N \to J/\psi N)$ of the `roberts2026` parametrisation against the photon energy on a nucleon at rest. Right: $d\sigma/dt$ at three photon energies; the forward limit $|t|_{\min}$ moves from 1.8 GeV^2 at 8.5 GeV to 0.36 GeV^2 at 10.5 GeV.

5 The generator and its validation

5.1 Algorithm

Each trial proceeds as follows.

1. E_γ is drawn from the spectrum (1) in $[k_{\min}, k_{\max}]$ by a tabulated inverse cumulative (20 001 points), including the endpoint factor.

2. The target nucleon is set at rest, or, on the deuteron, $\vec{p_s}$ is drawn from AV18, the struck four-vector and m^* are formed, and the trial is discarded if $m^* < 0.10$ GeV (0.01 % of the draws).

3. $s = (k + p_N)^2$ is computed; if it is below $(M_{J/\psi} + M_N)^2$ the trial is counted as below threshold and discarded.

4. The trial is accepted with probability $\sigma_{\mathrm{tot}}(s)\,(1 - \beta_{N,z})/w_{\max}$. $\sigma_{\mathrm{tot}}(s)$, the t integral of $d\sigma/dt$ over the card's range, is tabulated once on an 800-point grid in s. The majorant $w_{\max}$ is the maximum of $\sigma_{\mathrm{tot}}(s)\,(1 - \beta_{N,z})$ along the head-on line, a neutron running toward the photon with p_s up to the AV18 ceiling at $k_{\max}$: for a given $|p_s|$ both s and the flux factor are largest there, so this is a strict bound and no event is under-weighted (the generator counts overweights; there were none). The price is efficiency: $w_{\max}/\langle\sigma\rangle \simeq 36$ on the deuteron and 4 on a free nucleon.

5. t is drawn from the exact inverse cumulative of $d\sigma/dt$ at the event's s (401 points), ϕ uniformly.

6. The J/ψ and the recoil neutron are built in the γN centre of mass and boosted to the laboratory with the actual photon and nucleon four-vectors, so on the deuteron $k + p_D = p_{J/\psi} + p_{n'} + p_s$ exactly. The J/ψ is decayed to e^+e^- isotropically. The vertex is drawn along the cell as in Section 2.

The events are written in LUND format with the photon as the beam (pid 22, energy E_γ in the header), the target pid 2112 or 1000010020, the J/ψ as a status-0 parent with its daughters, the recoil and, on the deuteron, the spectator proton. The run's `config.json` records the photon count per electron in the generated range, the flux-weighted $\langle\sigma\rangle$, and the yield per incident electron per target nucleon, $Y = \int dE_\gamma \, (dN_\gamma/dE_\gamma) \, \langle(1-\beta_z)\sigma\rangle$.

5.2 Validation

- **Four-momentum conservation.** $|k + p_{\text{target}} - \sum p_{\text{final}}| < 5.5 \times 10^{-9}\,\text{GeV}$ for every event of the 200 000-event test samples, i.e. the ten significant digits of the LUND file; the spectator is on shell to 3×10^{-16}.
- **Photon spectrum.** On the free proton the generated E_γ distribution against the analytic $dN_\gamma/dE_\gamma \cdot \sigma_{\text{tot}}(W(E_\gamma))$: $\chi^2/\text{ndf} = 22.1/23$ over 6.5–10.6 GeV.
- t **slope.** An exponential fit of the generated t' distribution at $E_\gamma = 9.4$–$9.6\,\text{GeV}$ gives $b = 1.64\,\text{GeV}^{-2}$ against the model's 1.67 (the fit range includes the curvature of the physical limit).
- **Fermi kinematics.** The unweighted AV18 table of Table 1 is reproduced by the generator's own sampler; on the free neutron no event appears below 8.20 GeV, on the deuteron they do (Section 6.2).
- **Speed.** 0.09 ms per accepted event on a free nucleon and 0.65 ms on the deuteron on one core of the M4 Mac mini; the three 20-million-event samples took 35 minutes on ten cores, with no majorant violation in any of the 60 million events.

6 Results

6.1 Yield versus photon energy and the ratio deuteron / free neutron

Figure 3 shows the J/ψ yield per 100 MeV of photon energy, per incident electron and per neutron, on the deuteron and on a free neutron at rest, and their ratio above 8.5 GeV in 100 MeV bins. The per-electron normalisation is the same for both (same radiator, same dN_γ/dE_γ), so the ratio is the effect of Fermi motion and binding alone. With 20 million events per sample its statistical precision is 0.2 % per 100 MeV bin above 9 GeV.

Table 2: Yield per incident electron and per neutron (10^{-6} nb×photons/e) in bins of photon energy, on the deuteron and on the free neutron, their ratio with the statistical error, and the ratio obtained without the flux factor $1 - \beta_{N,z}$ (binding only).

E_γ [GeV]	deuteron	free neutron	ratio	ratio without flux factor
8.2–8.4	7.78	4.38	1.7757 ± 0.0047	1.6225
8.4–8.6	12.01	11.04	1.0879 ± 0.0020	1.0149
8.6–9.0	37.79	37.82	0.9992 ± 0.0010	0.9486
9.0–9.5	73.54	74.27	0.9902 ± 0.0007	0.9518
9.5–10.0	103.11	103.84	0.9929 ± 0.0006	0.9611
10.0–10.6	156.09	156.70	0.9961 ± 0.0005	0.9679
6.5–10.6 (all)	400.52	388.06	1.0321	0.9919

Table 2 gives the numbers. Above 8.6 GeV the ratio never leaves the band 0.99–1.00: it is 1.0173 ± 0.0023 in 8.6–8.7 GeV, passes through a shallow minimum of 0.9889 ± 0.0016 in 9.2–9.3 GeV, and comes back to 0.9961 in the last 600 MeV; its mean over 8.6–10.6 GeV is

$$\left.\frac{Y_d}{Y_n}\right|_{8.6<E_\gamma<10.6\,\text{GeV}} = 0.9944 \pm 0.0003. \tag{11}$$

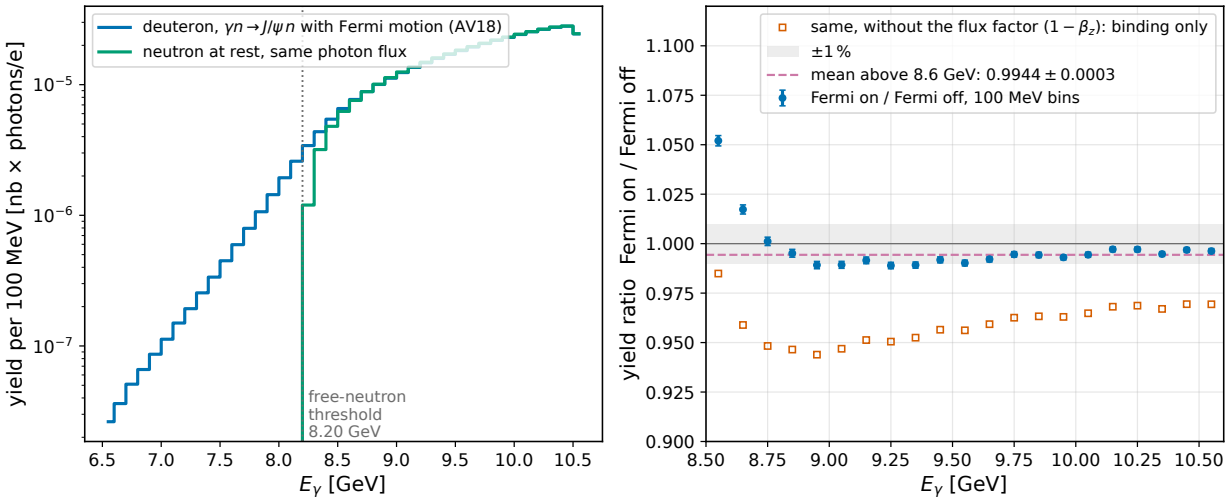


Figure 3: Left: J/ψ yield per 100 MeV of photon energy, per incident electron and per neutron, for the neutron in the deuteron (AV18) and for a free neutron at rest; the free threshold is at 8.20 GeV. Right: the ratio of the two above 8.5 GeV in 100 MeV bins (filled circles, statistical errors); the band is ±1 % and the dashed line the mean above 8.6 GeV. The open squares show the same ratio when the Møller flux factor $1-\beta_{N,z}$ of Eq. (9) is removed from the deuteron weights: this is the binding effect alone.

With 20 million events the statistical errors are small enough to resolve this sub-percent shape, which is why no straight line fits it; what matters is its size. The total yield per neutron over the whole generated range is 1.0321 times the free-neutron one; the excess is the near-threshold region, 8.2–8.6 GeV, and the sub-threshold yield of Section 6.2. **Above 8.6 GeV Fermi motion does not change the J/ψ yield per neutron by more than 1 %.** This is the net of two effects that the last column of Table 2 separates. Without the flux factor the ratio is 0.9611 at 9.5–10 GeV: the struck neutron carries less than its rest energy ($\langle E_N\rangle = M_D - \langle E_s\rangle = 0.933$ GeV against $M_n = 0.940$ GeV) and is off shell ($\langle m^*\rangle = 0.915$ GeV), so at fixed E_γ the average s is lower than on a free neutron by about 0.16 GeV^2, i.e. W by 18 MeV, and with $d\ln\sigma/dW \simeq 2.7\,\text{GeV}^{-1}$ the yield drops by 4–5 %. The flux-factor correlation of Section 3.1 adds back +4 %. The two have different energy dependence, both growing toward threshold, which is why the ratio without the flux factor rises slowly with E_γ from 0.9486 to 0.9679 while the full ratio stays within 1 % of one.

Near the free threshold the picture changes completely. In 8.4–8.6 GeV the ratio is 1.0879, in 8.2–8.4 GeV it is 1.7757 (2.6 in the first 100 MeV above threshold), and below 8.20 GeV the free-neutron yield is zero while the deuteron yield is not: Fermi motion smears the sharp threshold of $\sigma_{\text{tot}}(W) \propto (W - W_{\text{th}})^{0.7}$ over ±300 MeV of photon energy. Any extraction of the near-threshold cross section from deuterium data has to unfold this, and the generator provides the response.

6.2 The yield below the free threshold

Figure 4 shows the region below and around the free threshold. On the deuteron 2.55 % of the J/ψ yield in the generated range, 10.21×10^{-6} nb× photons per electron, is produced by photons below 8.20 GeV; 1.41 % below 8.0 GeV and 0.33 % below 7.5 GeV. The sample contains 13282, 52273, 216927 and 227595 events in 6.5–7.0, 7.0–7.5, 7.5–8.0 and 8.0–8.2 GeV. The yield falls by an order of magnitude per 500 MeV below threshold, following the high-momentum tail of $n(p_s)$ folded with the threshold rise of the cross section, and it depends on exactly the ingredients that are least certain: the AV18 tail beyond 300 MeV/c, the off-shell prescription and the absence of final-state interactions. The numbers below 8 GeV should be read as the plane-wave expectation, good to a factor of order two rather than to percent.

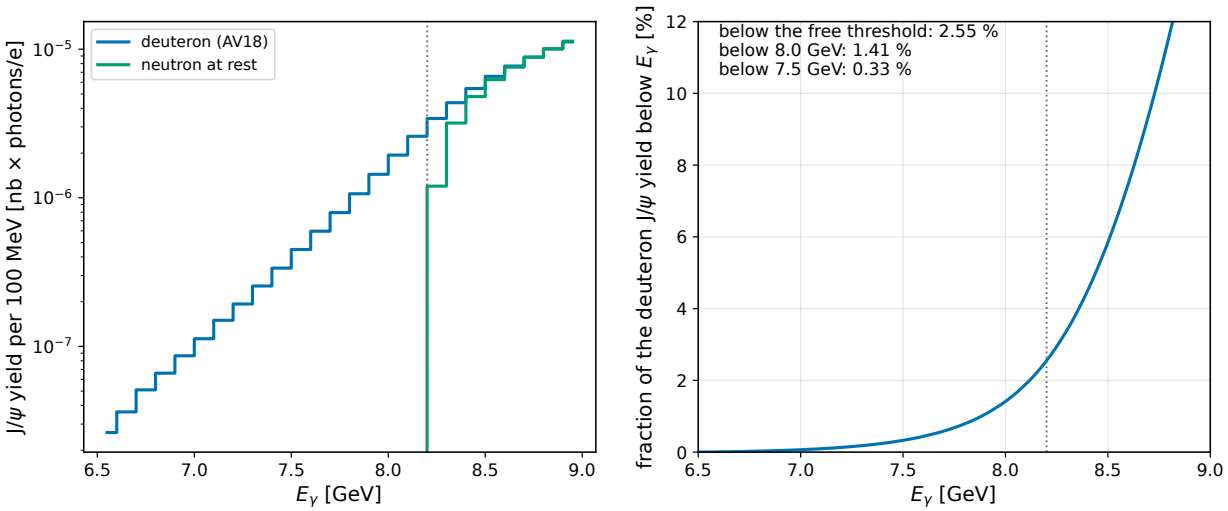


Figure 4: Left: the yield per 100 MeV between 6.5 and 9 GeV on the deuteron and on the free neutron (logarithmic scale). Right: cumulative fraction of the deuteron J/ψ yield below a given photon energy.

6.3 Spectator momenta and the off-shell mass

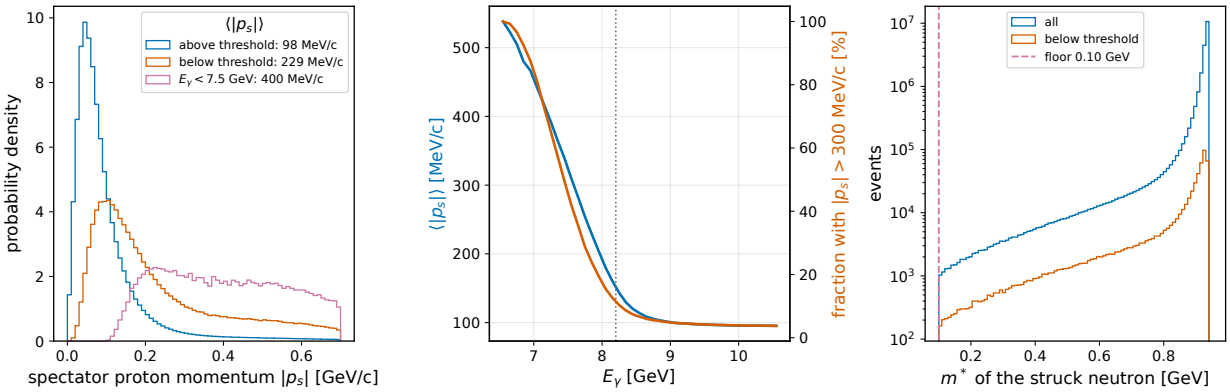


Figure 5: Left: distribution of the spectator proton momentum for events above the free threshold, below it, and below 7.5 GeV. Middle: mean spectator momentum and the fraction of events with $p_s > 300\,\mathrm{MeV}/c$ against the photon energy. Right: the off-shell mass m^* of the struck neutron, all events and sub-threshold events; the floor at 0.10 GeV is indicated.

The spectator proton is written in every deuteron event and Figure 5 shows what it looks like. Above the free threshold the spectators follow the bulk of AV18: $\langle p_s \rangle = 99$ MeV/c and 4.4 % above 300 MeV/c, with $\langle m^* \rangle = 0.915$ GeV. Below threshold the events are selected from the tail: $\langle p_s \rangle = 229\,\mathrm{MeV}/c$ and 26.2 % above 300 MeV/c for all sub-threshold events, 278 MeV/c and 36.2 % below 8.0 GeV, 400 MeV/c and 67.6 % below 7.5 GeV, with $\langle m^* \rangle$ falling to 0.706 GeV. The floor at 0.10 GeV removes 0.01 % of the draws overall and is invisible in any of these distributions. A sub-threshold J/ψ comes with a fast spectator proton at large angle that CLAS12 can detect, which is the experimental handle on this part of the sample.

6.4 The t distributions

Figure 6 shows $-t = -(k - p_{J/\psi})^2$, computed as an analysis would from the photon and the J/ψ, without knowledge of the spectator. Above threshold the deuteron and the free-nucleon distributions coincide: the t dependence is generated at the event's own s and Fermi motion only redistributes events among nearby s. Below the free threshold the events sit at the edge of phase space, $|t|_{\min}$ is large and the distribution is compressed toward it; the t slope of a sub-threshold

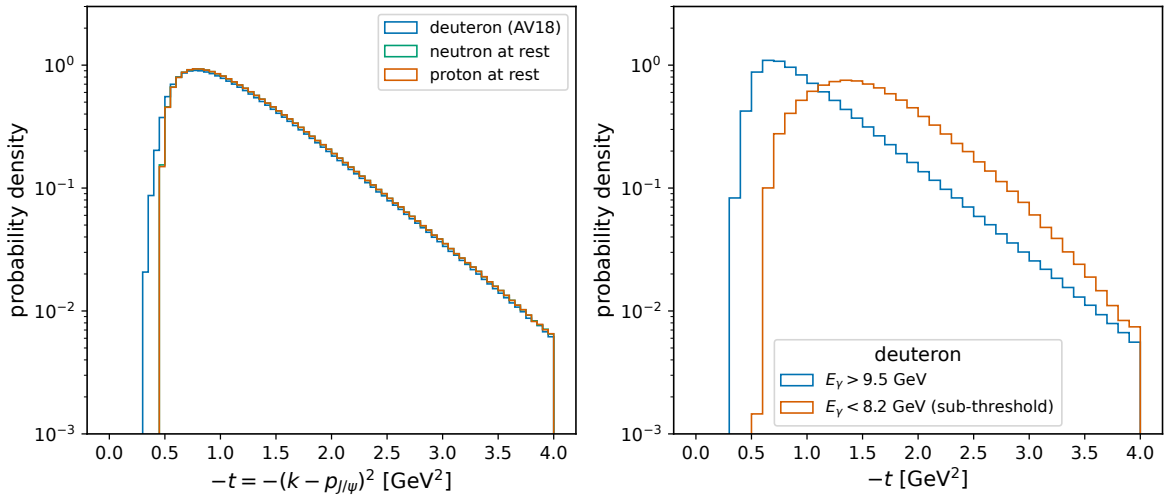


Figure 6: Left: $-t = -(k - p_{J/\psi})^2$, with k the photon four-vector, for the three samples. Right: on the deuteron, above 9.5 GeV and below the free threshold.

sample is not the elementary slope b.

7 Discussion and limits

What the result means. For the J/ψ yield per neutron above 8.6 GeV, a generator without Fermi motion is as good as one with it, to better than 1 % at every photon energy. This is not because Fermi motion is small (the smearing of s is $\pm 0.7\,\mathrm{GeV}^2$) but because its two first-order effects on a steeply rising cross section, binding and the flux-factor correlation, cancel to that level for the AV18 deuteron. The cancellation is specific to the observable (yield at fixed lab photon energy, integrated over the spectator) and to the steepness of $\sigma(W)$; it would not hold for a different α or for a target with larger binding.

Where Fermi motion is essential. Within 400 MeV of the free threshold the ratio departs from one by tens of percent and below 8.20 GeV the whole yield is a Fermi-motion effect. The E_γ-dependence of a threshold measurement on deuterium must be unfolded with this response, and the spectator proton, when detected, tags the events that carry it.

Limits of the model. The calculation is plane-wave impulse approximation with a spectator on shell and the cross section evaluated at the free-nucleon mass. Not included: final-state interactions of the J/ψ or the recoil neutron with the spectator, coherent production on the deuteron, any modification of the elementary amplitude off shell, the D-wave tensor structure (only $|\psi|^2$ enters), and the difference between σ_n and σ_p. The AV18 tail above 300 MeV/c is used as is. The bremsstrahlung is thin-target and does not include the energy loss before radiation or the beam energy spread, each of order 1 MeV at the endpoint. The elementary cross section is one parametrisation, `roberts2026`, with $b = 1.67\,\mathrm{GeV}^{-2}$ independent of W; the ratio of Section 6.1 depends on its slope near threshold, not on its normalisation.

A Cards, commands and files

The deuteron card (`photo20M/card_d2.txt`); the neutron and proton cards differ only in the model (`jpsin.roberts2026` with `fermi = none`; `jpsi.roberts2026`), in `target_X0_cm = 890.4` for LH_2 and in the output directory:

```
model = jpsin.roberts2026
```

```
ebeam = 10.6
kmin  = 6.5
kmax  = 10.6
tmin  = -6.0
tmax  = -0.001
nev   = 20000000
seed  = 2027
fermi = av18
fermi_mstar_min = 0.10
target_length_cm = 5.0
target_X0_cm = 769.1
vz = -5.5 -0.5
idecay = 1
maj_safety = 1.05
outroot = ROOT/photo20M/d2
```

Generation, figures and this document:

```
python3 -m exclurad_py.generator.photo card_d2.txt --jobs 10
python3 validation/jpsi_photo_report_figs.py ROOT/photo20M ROOT/photo20M/report
python3 validation/numbers_to_tex.py ROOT/photo20M/report/numbers.json numbers.tex
pdflatex jpsi_fermi_photoproduction.tex
```

The samples (LUND files, 42 GB) are on the Mac mini, with `ROOT` the directory `/Volumes/wd_14tb/aao_norad_kin/jpsi_fermi`; each run directory carries its `inputs/config.json`. The figures, `numbers.json` and this report are in the `jpsi-fermi` branch under `docs/jpsi_fermi/report/`, and in the OneDrive folder `Work/2026_Jpsi_Fermi_photoproduction`.